

Final Review

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August 7, 2026

Novel Prep

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Phase 3 · Final Review

Overview

1 Final Review

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11 Phase 3 final review questions

Question 1

In the given equation, k is a constant. The equation has exactly one real solution. What is the value of k ?

$$x^2 + (\sqrt{k-3})x + 42 = 0$$

- A. 171
- B. 168
- C. 165
- D. 45

Answer 1

Correct Answer: A

Question 2

In the given system of equations, m and b are negative constants. In the xy -plane, the graphs of the equations in the given system intersect at the point $(-5, y)$, where $y < 0$. Which expression represents the value of b ?

$$x^2 + y^2 = 36$$

$$y = mx + \frac{b}{4}$$

A. $-\frac{5m}{4} + \frac{\sqrt{11}}{4}$

B. $\frac{5m}{4} - \frac{\sqrt{11}}{4}$

C. $-20m + 4\sqrt{11}$

D. $20m - 4\sqrt{11}$

Answer 2

Correct Answer: D

Question 3

A biologist mixed a solution that is 0.3% sodium chloride by mass with a solution that is 0.15% sodium chloride by mass to obtain a new solution, which has a mass of 80 grams and contains 0.21 grams of sodium chloride. How many grams of 0.3% sodium chloride solution did the biologist use?

- A. 0.14
- B. 20
- C. 60
- D. 79.86

Answer 3

Correct Answer: C

Question 4

The function v is defined by the equation

$$v(x) = \frac{x^2 + bx + c}{(x + 5)(x - 19)},$$

where b and c are constants. In the xy -plane, the graph of $y = v(x)$ passes through the points $(0, \frac{66}{95})$ and $(11, 0)$. If $v(q) = 0$, which of the following could be the value of q ?

- A. -6
- B. -5
- C. 5
- D. 6

Answer 4

Correct Answer: A

Question 5

$$\frac{x}{5} + \frac{y}{9} = \frac{47}{45}$$

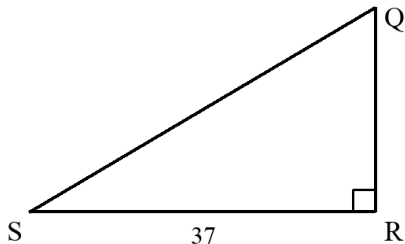
An engineer connects resistors in series, where the resistors in the series have a total resistance of $\frac{47}{45}$ ohms. In this series, there are x resistors of type A, which each have a resistance of a ohms, and y resistors of type B, which each have a resistance of b ohms. The given equation represents this situation. According to this equation, what is the positive difference between the value of a and the value of b ?

- A. 47
- B. 4
- C. $\frac{47}{45}$
- D. $\frac{4}{45}$

Answer 5

Correct Answer: D

Question 6



Note: Figure not drawn to scale.

In triangle QRS shown, $QR < RS$. Which expression represents the length of $\overline{QS}$?

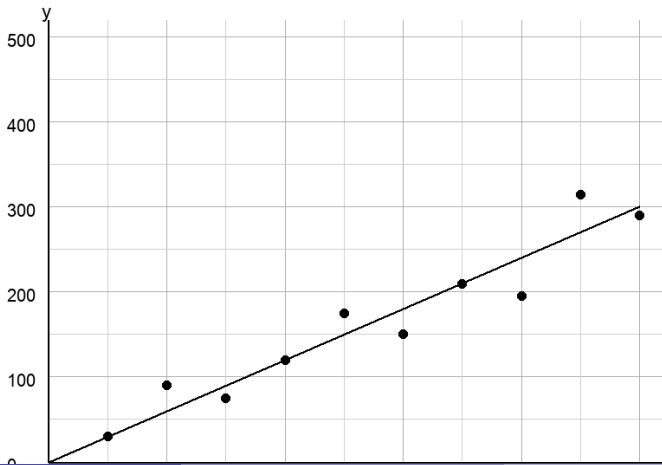
- A. $37 \cos Q$
- B. $37 \sin Q$

Answer 6

Correct Answer: D

Question 7

A group of 10 gardeners recorded data on the germination rates of their tomato crop for one growing season. The scatterplot shows the relationship between the number of tomato seeds planted, x , and the number of tomato seeds that germinated, y , for each of the gardeners. A line of best fit is also shown.



Answer 7

Correct Answer: C

Question 8

A rectangular banner has an area of 3,200 square inches. A copy of the banner is made in which the length and width of the original banner are each increased by 20%. What is the area of the copy, in square inches?

- A. 3,220
- B. 3,240
- C. 3,840
- D. 4,608

Answer 8

Correct Answer: D

Question 9

The mass of object A is 444% of the mass of object B, and the mass of object A is 0.740% of the mass of object C. If the mass of object C is $p\%$ of the mass of object B, what is the value of p ?

Answer 9

Final Answer:

Question 10

In 2013, Chloe earned 11% more than in 2012, and in 2014 Chloe earned 6% more than in 2013. If Chloe earned y times as much in 2012 as in 2014, which of the following is closest to the value of y ?

- A. 0.5455
- B. 0.6600
- C. 0.8499
- D. 1.1766

Answer 10

Correct Answer: C

Question 11

Point F lies on a unit circle in the xy -plane and has coordinates $(1, 0)$. Point G is the center of the circle and has coordinates $(0, 0)$. Point H also lies on the circle and has coordinates $(-1, y)$, where y is a constant. Which of the following could be the positive measure of angle FGH , in radians?

- A. $\frac{37\pi}{2}$
- B. $\frac{39\pi}{2}$
- C. 34π
- D. 35π

Answer 11

Correct Answer: D

Final Advice

Keep these habits with you after today.

01 Desmos is a tool, not a crutch

Use Desmos for graphs, points, or quick arithmetic — not for every question. Many items are faster and safer with algebra or a short written setup. Solve by hand when you can; use Desmos to verify when unsure.

02 Read the ask first, then harvest the stem

Start with the last line: what is the question asking? Then go line by line and ask, “What math idea is this pointing to?” Each line often hints at a formula, ratio, or constraint. Capture the hints, choose a method, then compute.

03 Build speed without giving up accuracy

Spend less time stuck on one item. Recognize the type and skill quickly. If you are spinning, mark it, move on, and return later. Accuracy first — then faster recognition.

FROM HERE

How to Prep

Short, focused practice beats a long vague season.

04 Keep prep focused

Use a short, concentrated block: drills, mocks, and review of misses. You already have enough Novel Prep questions for SAT Math — find gaps and fix them.

05 Close the loop

After each set, write 2–3 skills you missed. Drill those next — not the whole book again.

06 Manage the clock

Skip and return. Do not burn four minutes early on one hard problem. Easy and medium points first.

07 Trust your method

Restate the ask, underline givens, eliminate choices, or plug in a number. The habits from class are enough — use them calmly.

Short prep. Clear method. Smart use of tools. You've got this.

Thank You!

We appreciate your time and focus.

Novel Prep